

COVER PAGE

Title: Python Basics – Week 1 Curriculum & Teaching Portfolio

Prepared By: ATHIRA A

Subject: Python Programming for Beginners

Target Audience: Ages 10+, School Students & Adult Beginners

Duration: 1 Week (5 Days)

Tools Used: Python (Replit / IDLE / Online Compiler / Code.org)

Date: 19-11-2025

CURRICULUM OUTLINE (WEEK 1)

Day 1 – Introduction to Python & Printing Output

Learning Objectives:

- Understand what programming is
- Learn where Python is used
- Use print()
- Display text and numbers

Activities:

- Icebreaker
- Demo: First Python code
- Hands-on: 5 print statements

Outcome:

Students can write simple programs that display messages.

Day 2 – Variables & Data Types

Learning Objectives:

- Understand variables
- Store text & numbers
- Know int, float, string

Activities:

- Explain variables using real-life examples
- Demo with variables
- Practice tasks

Outcome:

Students can create and print variables.

Day 3 – Input & Basic Expressions

Learning Objectives:

- Use input ()
- Perform basic operations
- Convert types

Activities:

- Demo: Taking name & age
- Arithmetic practice
- Mini calculator

Outcome:

Students can build programs with input and expressions.

Day 4 – Conditional Statements (if/else)

Learning Objectives:

- Understand conditions
- Compare numbers & strings
- Use if/else

Activities:

- Age check program
- Grading system
- Login simulation

Outcome:

Students understand decision-making in Python.

Day 5 – Project Day

Students choose one:

- Age Calculator
- Number Comparison Tool
- Login System
- Simple Chat Responder

Outcome:

Applies concepts from Days 1–4.

Detailed Lesson Plan:

DAY 1-INTRODUCTION TO PYTHON AND PRINTING OUTPUT

Lesson Title: Introduction to Python & Printing Output

Grade Level: Beginner (Age 10+ / Coding Beginners)

Duration: 45–60 minutes

1. Learning Objectives

By the end of this lesson, students will be able to:

- Describe what a programming language does.
- Understand what Python is and where it is used.
- Write their first Python program using the print () function.
- Print text, numbers, and simple expressions in Python.
- Identify common printing errors (missing quotes, parentheses).

2. Materials Needed

- Laptop or desktop with internet access
- Python IDE (Replit / IDLE / Code.org Python Lab)

3. Lesson Introduction (10 minutes)

"Imagine if your computer could understand your instructions exactly like a smart friend. Python is a language that allows us to talk to the computer!"

Explain:

- Python is a programming language used for apps, websites, AI, and automation.
- It is easy to read and great for beginners.
- Today, students will create their first Python program.

Real-life examples of Python usage:

- Instagram backend
- Google search algorithms (parts)
- Robotics & automation
- Data science and AI models

4. Direct Instruction (15 minutes)

4.1 What is print ()?

- Python uses the print () function to show output on the screen.
- Think of it as asking the computer to "say something".

Syntax:

```
print ("Your message here")
```

4.2 How to Print Text

- Text must be inside quotes.

Example:

```
print ("Hello, Python!")
```

Output:

Hello, Python!

4.3 How to Print Numbers

- Numbers do not need quotes.

Example:

```
print (25)
```

4.4 Printing Multiple Items

```
print ("My age is", 12)
```

4.5 Why Quotes Matter

Explain the difference:

```
print ("20") # Text
```

```
print (20) # Number
```

4.6 Common Errors

Missing Quotes:

```
print (Hello)
```

Error: `NameError`

Missing Parenthesis:

```
print "Hello"
```

Error: `SyntaxError`

5. Guided Practice (10 minutes)

Students try the following with teacher support:

1. Print their name

```
print ("My name is Ann")
```

2. Print their favourite food

```
print ("I love biryani")
```

3. Print a number

```
print (2025)
```

4. Print a sentence with text + number

```
print ("I am", 20, "years old")
```

The teacher checks and corrects errors.

6. Independent Practice (10 minutes)

Students complete small tasks:

- Task 1: Print 3 sentences about yourself.

- Task 2: Print your favourite hobby.
- Task 3: Print any number between 1–100.
- Task 4: Print this exact output:

Python is fun!

7. Mini Project (5 minutes)

Create a Personal Introduction Program.

Students must include:

- Name
- Age
- One favourite thing
- One dream for the future
- A number of their choice

Sample Output:

My name is Rahul.

I am 12 years old.

I love football.

My dream is to become a scientist.

My favourite number is 18.

8. Assessment / Exit Ticket (5 minutes)

Ask students to answer:

1. What does the print () function do?
2. Why do we use quotes?
3. Write a correct Python print statement for:

9. Homework

Create a Python script with 5 print statements describing:

- Your best friend
- A place you want to visit
- A number fact
- A motivational quote
- Anything funny

10. Teacher Notes / Tips

- Encourage students to try different variations of print.
- Remind them: text → quotes, numbers → no quotes.
- Ask them to read error messages — learning to debug is important.
- Celebrate their first successful program.

MCQ Quiz + Answer Key

1. What does the print () function do in Python?

- a) Adds two numbers
- b) Shows output on the screen
- c) Deletes text
- d) Saves the program

Ans: b) Shows output on the screen

2. Which of the following is the correct way to print text?

- a) print (Hello)
- b) print ("Hello")
- c) print ("Hello)
- d) print (Hello")

Ans: b) print("Hello")

3. Which of these does NOT need quotes?

- a) "Hello"
- b) "Python"
- c) 25
- d) "2025"

Ans: c) 25

4. What error occurs when you forget quotes around text?

- a) ZeroDivisionError
- b) NameError
- c) TypeError
- d) ValueError

Ans: b) NameError

5. Which command will print the number 10?

- a) print("10")
- b) print(10)
- c) print(ten)

d) print("ten")

Ans: b) print (10)

DETAILED LESSON PLAN (DAY 2 – VARIABLES & DATA TYPES)

Lesson Title: Variables & Data Types

Duration: 45–60 minutes

Materials: Laptop, Replit/Compiler

Learning Objectives

Students will:

- Define a variable
- Create variables
- Identify string, int, float
- Print values
- Write simple programs

Lesson Flow

1. Warm-Up – 5 min

Ask: “Where do we store things in real life?”

Explain how variables store information in Python.

2. Introduction to Variables – 10 min

- Variable = name that holds a value

Examples:

name = "John"

age = 12

3. Data Types – 10 min

- String – text
- Integer – whole number
- Float – decimal

Examples:

city = "Mumbai"

height = 160

temperature = 36.6

4. Hands-On Coding – 15 min

Students create:

fruit = "Apple"

quantity = 5

price = 12.50

print(fruit)

print(quantity)

print(price)

5. Challenge Activity – 10 min

Students create a personal info program.

Assessment (Exit Ticket)

- What is a variable?
- Example of a string variable
- Create a float variable
- Predict output

Homework

Create a Python program that stores the following information using variables:

Your favorite movie

The year it was released

The main actor's name

The rating you would give (out of 10, using a float)

Then print all four values.

MCQ QUIZ (10 Questions)

Q1. What is Python?

- A. A type of snake
- B. A programming language
- C. A computer
- D. None

Answer: B

Q2. Which function prints output?

- A. show ()

B. output ()

C. print ()

D. display ()

Answer: C

Q3. Which is a string?

A. 12

B. 4.5

C. "Hello"

D. -7

Answer: C

Q4. Which is an integer?

A. "14"

B. 3.14

C. 14

D. "3.14"

Answer: C

Q5. What does input () do?

A. Shows output

B. Takes user input

C. Adds numbers

D. Stops program

Answer: B

Q6. Which symbol is used for multiplication?

A. x

B. *

C. /

D. –

Answer: B

Q7. What does this code do?

```
age = 10
```

```
print(age)
```

A. Prints 10

B. Prints age

C. Error

D. Prints nothing

Answer: A

Q8. float refers to:

A. Whole numbers

B. Decimal numbers

C. Text

D. True or False

Answer: B

Q9. Which is a correct variable?

A. 1name = "Ann"

B. @age = 10

C. name = "Ann"

D. #city = "Delhi"

Answer: C

Q10. If $x = 5$ and $y = 3$, what is $x + y$?

A. 8

B. 2

C. 53

D. Error

Answer: A

Day 3 – Input & Basic Expressions

Lesson Title: Input & Basic Expressions

Duration: 45–60 minutes

Materials: Laptop, Replit/Compiler

Learning Objectives

By the end of this lesson, students will be able to:

- Use the input () function to take user input.
- Perform basic arithmetic operations (+, −, *, /, //, %, **).
- Convert data types (e.g., int (), float (), str ()).
- Build small interactive programs using input and expressions.

1. Warm-Up – 5 min

Ask students: “How do websites or apps know your name or age?”

Explain: Programs ask users for information using the input () function.

2. Introduction to Input – 10 min

Concept: input () lets the user type something during the program.

It always returns a string.

Example:

```
name = input ("Enter your name: ")
```

```
print ("Hello", name)
```

Discuss how input makes programs interactive.

3. Basic Expressions & Type Conversion – 10 min

Introduce operators:

- Addition (+)
- Subtraction (-)
- Multiplication (*)
- Division (/)
- Floor Division (//)
- Modulus (%)
- Exponent (**)

Explain type conversion: int (), float (), str ()

Example:

```
age = int (input ("Enter your age: "))
```

```
years = age + 5
```



```
print ("You will be", years, "in 5 years.")
```

4. Hands-On Coding – 15 min

Students create a small program using input and operators:

```
num1 = float (input ("Enter first number: "))
```

```
um2 = float (input ("Enter second number: "))
```

```
print ("Sum =", num1 + num2)
```

```
print ("Difference =", num1 - num2)
```

```
print ("Product =", num1 * num2)
```

```
print ("Division =", num1 / num2)
```

The teacher checks if students used the correct type conversion.

5. Challenge Activity – 10 min

Task: Build a Mini Calculator that displays:

- Addition
- Subtraction
- Multiplication
- Division
- Remainder (%)

Students must use input + operators + conversion.

Assessment (Exit Ticket)

1. What does the input () function do?
2. Why do we use int () or float ()?

3. What is the output of: `print (10 % 3)`?

4. Predict the output of this program:

```
x = int (input ("Enter number: "))
```

```
print (x * 2)
```

Homework

Create a program that asks the user for two numbers and prints:

- Their sum
- Their average
- The larger number Use input, operators, and type conversion.

MCQ QUIZ (10 Questions)

1. What does the `input ()` function return by default?

- a) Integer
- b) Float
- c) String
- d) Boolean

Answer: c

2. Which function is used to convert a string to an integer?

- a) `str()`
- b) `float()`
- c) `int()`

d) input ()

Answer: c

3. What is the output of: `int ("5") + int ("3")`?

a) 53

b) 8

c) 5+3

d) Error

Answer: b

4. Which operator performs exponentiation in Python?

a) ^

b) **

c) *

d) //

Answer: b

5. What will `10 // 3` return?

a) 3.33

b) 3

c) 4

d) 0

Answer: b

6. Which of the following is correct for taking a number as input?

- a) `num = input (int ("Enter a number: "))`
- b) `num = int (input ("Enter a number: "))`
- c) `num = input ("Enter a number:", int)`
- d) `num = number (input ())`

Answer: b

7. What will be the output of: `print (7 % 3)`?

- a) 1
- b) 2
- c) 3
- d) 4

Answer: a

8. Which of these is NOT an arithmetic operator?

- a) +
- b) %
- c) //
- d) &&

Answer: d

9. What happens if you add a string and an integer?

- a) They add normally
- b) Python converts automatically
- c) Error: Type mismatch
- d) It prints them together

Answer: c

10. Which statement correctly prints the user's name?

- a) `print ("Hello" + name)`
- b) `print ("Hello", name)`
- c) Both a and b
- d) None

Answer: c

Day 4 – Conditional Statements (if/else)

Lesson Title: Conditional Statements (if/else)

Duration: 45–60 minutes

Materials: Laptop, Replit/Compiler

Learning Objectives

By the end of this lesson, students will be able to:

- Understand what a condition is
- Compare numbers and strings using comparison operators
- Use if and else to make decisions in programs

Lesson Flow

1. Warm-Up – 5 min

Ask students:

“Have you ever chosen ‘If it rains, I will take an umbrella’?”

Explain that computers also make decisions using if/else statements.

2. Introduction to Conditions – 10 min

Explain comparison operators:

- `==` equal to
- `!=` not equal to
- `>` greater than
- `<` less than
- `>=` greater than or equal
- `<=` less than or equal

Example:

```
Print (5 > 3) # True
```

```
Print ("A" == "A") # True
```

```
Print (10 != 7) # True
```

3. Using if/else – 10 min

Explain structure:

if condition:

```
    # code runs if True
```

else:

```
    # code runs if False
```

Introduce elif, indentation, and its usage.

Example:

```
age = 15
```

```
if age >= 18:
    print ("You can vote.")
else:
    print ("You cannot vote yet.")
```

4. Hands-On Coding – 15 min

Students practice the following programs:

a) Age Check Program

```
age = int (input ("Enter your age: "))
```

```
if age >= 18:
    print ("You are an adult.")
else:
    print ("You are a minor.")
```

b) Grading System

```
marks = int (input ("Enter your marks: "))
```

```
if marks >= 90:
    print ("Grade A")
else:
    print ("Keep trying!")
```

c) Login Simulation

```
username = input ("Enter username: ")
```

```
if username == "admin":
```

```
    print ("Login successful")
else:
    print ("Wrong username")
```

5. Challenge Activity – 10 min

Task:

Create a program that checks whether a number is:

- Positive
- Negative
- Zero

Students must use if, elif, and else.

Assessment (Exit Ticket)

1. What is a condition?
2. Write an example of an if/else statement.
3. What is the output of:

```
    if 5 > 10:
        print("Yes")
```

```
    else:
        print("No")
```

4. Compare two strings using ==.

Homework

Create a Python program using if/else to check whether a student has passed or failed.

- Input: marks

- Condition: marks $\geq 40 \rightarrow$ Pass
- Else $\rightarrow$ Fail

10 MCQs for Conditional Statements (if/else)

1. Which symbol is used for equality comparison in Python?

- a) =
- b) ==
- c) ===
- d) !=

Answer: b

2. What will this code print?

```
if 10 > 5:
```

```
    print("Yes")
```

```
else:
```

```
    print("No")
```

- a) Yes
- b) No
- c) Error
- d) Nothing

Answer: a

3. Which of the following is a valid if statement?

- a) if x > 3:
- b) if (x > 3)

c) if $x > 3$ then

d) if $x > 3$;

Answer: a

4. What does the else part do?

a) Runs only if the condition is true

b) Runs only if the condition is false

c) Runs only when there is an error

d) Runs before the if condition

Answer: b

5. What is the output of:

```
if "A" == "a":
```

```
    print("Match")
```

```
else:
```

```
    print ("No Match")
```

a) Match

b) No Match

c) Error

d) a

Answer: b

6. Which operator checks “not equal to”?

a) =!

b) !=

c) !=

d) <>

Answer: c

7. Identify the correct structure:

a)

if condition:

```
print("Hello")
```

b)

if condition

```
    print("Hello")
```

c)

if condition:

```
    print("Hello")
```

d)

if (condition):

```
print("Hello")
```

Answer: c

8. What will this code print?

```
age = 12
```

```
if age >= 18:
```

```
    print("Adult")
```

```
else:
```

```
    print("Child")
```

a) Adult

b) Child

c) Nothing

d) Error

Answer: b

9. What type of value does a condition return?

a) int

b) string

c) boolean (True/False)

d) float

Answer: c

10. Which statement is true about if/else?

a) else must always come before if

b) if can exist without else

c) A program can only have one if statement

d) if/else only works with numbers

Answer: b

Day 5 – Project Day

Theme: Apply all concepts learned from Days 1–4

(Input, Variables, Data Types, Expressions, if/else)

Lesson Flow

1. Warm-Up – 5 min

Ask students:

“What kind of programs do you want to build in real life?”

Explain that today they will choose and build their own mini-project.

2. Project Options – 5 min

Students choose one of the following projects:

Option A: Age Calculator

Use input, variables, expressions:

```
birthyear = int (input ("Enter your birth year: "))
```

```
age = 2025 - birthyear
```

```
print ("Your age is:", age)
```

Option B: Number Comparison Tool

Use input, type conversion, if/else:

```
a = int (input ("Enter first number: "))
```

```
b = int (input ("Enter second number: "))
```

```
if a > b:
```

```
    print ("First number is larger")
```

```
elif b > a:
```

```
    print ("Second number is larger")
```

```
else:
```

```
    print ("Both are equal")
```

Option C: Login System

Use strings and if/else:

```
username = input ("Enter username: ")
```

```
if username == "admin":
```

```
    print ("Welcome Admin!")
```

```
else:
```

```
    print ("Access Denied")
```

Option D: Simple Chat Responder

Use input + conditional logic:

```
message = input ("How are you? ")
```

```
if message == "good":
```

```
    print ("That's great!")
```

```
else:
```

```
    print ("Hope your day gets better!")
```

3. Work Time – 20 min

Students code their selected project individually or in pairs.

The teacher helps with debugging and logic.

4. Showcase – 10 min

Students run their program and explain:

- Which project did they choose

- What inputs does it take
- How the logic works

5. Learning Outcome

By the end of Day 5, students can:

- Use input correctly
- Declare variables
- Perform calculations
- Make decisions using if/else
- Combine all these skills to create a working mini-project

6. Assessment (Exit Ticket)

- What project did you choose?
- Which Python concept was hardest for you?
- What would you add if you had more time?

7. Homework

Improve today's project by adding one extra feature.

Examples:

- Age calculator → also shows future age
- Login system → add password
- Chat responder → add more responses
- Number comparison → also show difference